

Name:		Centre/Index Number:		Class:	
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H3 CHEMISTRY

Paper 1

9813/01

24 September 2021

2 hours 30 minutes

Candidates answer on the Question Paper.

Additional Materials: Data Booklet
 Insert

INSTRUCTIONS TO CANDIDATES

- 1 Write your name, centre number, index number and class at the top of this page.
- 2 Write in dark blue or black pen.
- 3 You may use a HB pencil for any diagrams or graphs.
- 4 Do not use staples, paper clips, glue or correction fluid.
- 5 Answer **all** questions in the spaces provided on the Question Paper. If additional space is required, you should use the blank pages in the booklet. The question number must be clearly shown.

Section A

Answer **all** questions.

Section B

Answer **two** questions.

The use of an approved scientific calculator is expected, where appropriate.

The number of marks is given in brackets [] at the end of each question or part question.

You are advised to show all workings in calculations.

You are reminded of the need for good English and clear presentation in your answers.

For Examiner's Use	
1	15
2	10
3	11
4	15
5	9
6	20
7	20
8	20
Overall	100

Section A

Answer **all** the questions in this section.

The information provided in the insert is taken from published scientific articles.

You should read the whole insert before you start to answer any questions and use the information it contains to answer the questions.

- 1 (a) (i) Suggest a potential benefit of using Lumiflavin as a photoredox catalyst over the use of either an iridium-based or ruthenium-based catalyst. [1]
- (ii) Suggest a reason why a C–H bond needs to be activated. [1]
- (iii) A Kessil lamp is specially designed to enable scientists to achieve optimal photoredox reactions.

Use the VMFP bioconjugation reaction conditions in Abstract 1 and data from the *Data Booklet* to calculate the HOMO–LUMO energy gap of Lumiflavin. [2]

- (iv) Using the colour wheel below, predict and explain the colour of Lumiflavin.



[2]

- (v) Step 1 in Fig. 1.1 involves the single electron transfer between the excited state of Lumiflavin, ^3LF , and methionine.

Using the reduction potentials provided in Table 1.1, calculate the standard Gibbs free energy change, $\Delta G^\ominus$, for the reaction.

Table 1.1

Electrode reaction	$E^\ominus / \text{V}$
$^3\text{LF} + \text{e}^- \rightleftharpoons \text{LF}^-$	+1.50
Radical cationic methionine + $\text{e}^- \rightleftharpoons$ methionine	+1.36

[2]

- (vi) State the type of reaction that occurred in Step 3 of Fig 1.1. [1]

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Suggest why hydrochloric acid is **not** a suitable proton source.

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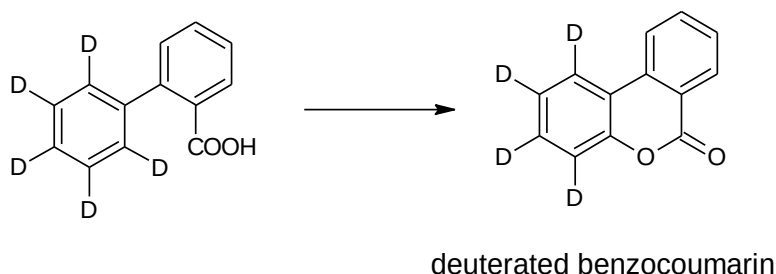
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- (c) The kinetic isotope effect (KIE) is a mechanistic phenomenon wherein isotopically substituted molecules react at different rates. The deuterium KIE is the ratio of the rate constant for a compound containing hydrogen, k_H , to the rate constant observed for an identical compound with one or more hydrogen atoms substituted with deuterium atoms, k_D .

$$\text{deuterium kinetic isotope effect} = \frac{k_H}{k_D}$$

The rate constant and hence the overall rate of reaction will differ if the C–H bond is broken in the rate determining step.

Gonzalez-Gomez et al carried out KIE experiments to study the lactonisation reaction found in Abstract 2. The concentration of benzocoumarin and deuterated benzocoumarin formed were measured at timed intervals.



Their results are shown in Tables 1.2 and 1.3.

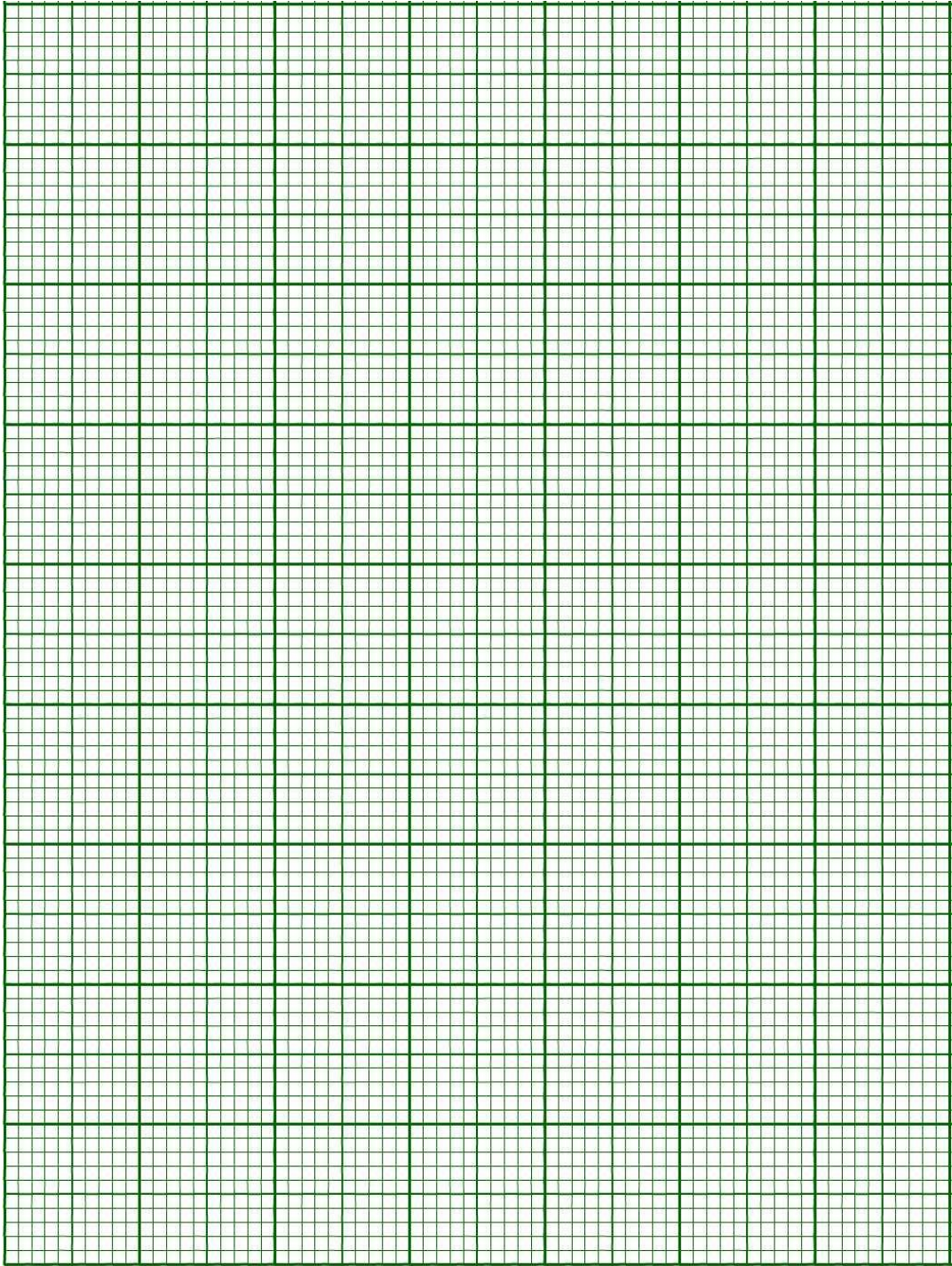
Table 1.2

t / min	Concentration of benzocoumarin / mmol dm ⁻³
10	0.96
33	4.48
54	7.71

Table 1.3

t / min	Concentration of deuterated benzocoumarin / mmol dm ⁻³
11	1.01
28	2.80
57	6.16

- (i) Using the data in Tables 1.2 and 1.3, draw suitable graphs on the grid given in page 5 to determine the value of $\frac{\text{rate of formation of benzocoumarin}}{\text{rate of formation of deuterated benzocoumarin}}$ which is also the $\frac{k_H}{k_D}$ ratio for the lactonisation reaction. [4]
- (ii) The $\frac{k_H}{k_D}$ ratio of E2 mechanism is generally 6–7 while the $\frac{k_H}{k_D}$ ratio of E1 mechanism is generally 1–2. Use these literature values and your answer to (c)(i) to deduce if the C–H bond is broken in the rate-determining step of the lactonisation reaction. [1]



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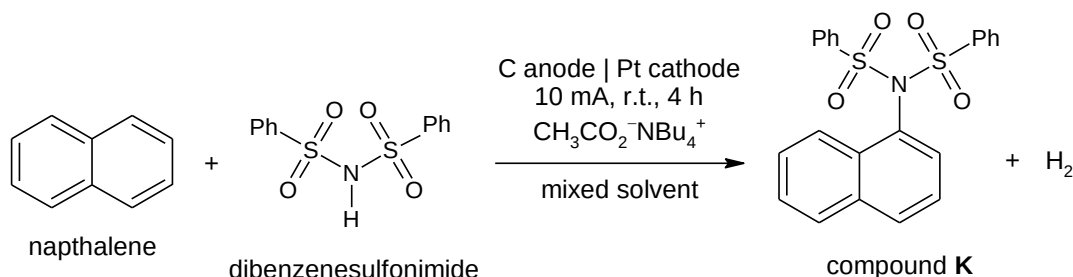
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[Total: 15]

- 2 (a) In 2019, Lei and co-workers reported an electrochemical method to install nitrogen functionalities onto aromatic compounds.

An example is the synthesis of compound **K** from naphthalene shown in Fig. 2.1.



where Ph = phenyl, C_6H_5 and Bu = butyl, $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$

Fig. 2.1

The first two steps of the mechanism are described as follows.

- step 1 Homolytic fission of the N–H bond of dibenzenesulfonimide generates a radical intermediate and a proton with the loss of a single electron at the anode.
- step 2 Radical addition of the intermediate in step 1 with naphthalene forms a C–N bond and another radical intermediate.

- (i) Construct a balanced equation for step 1 of the mechanism. [1]
- (ii) Complete Fig. 2.2 with curly arrows to show the movement of electrons for the radical addition occurring in step 2. [2]

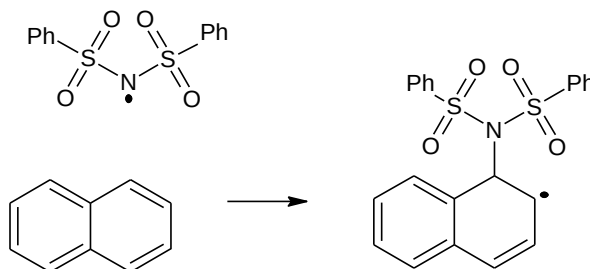


Fig. 2.2

- (iii) By applying the Hammond postulate, determine whether the transition state for step 2 resembles the reactants or product more. [1]

Quote relevant data from the *Data Booklet* to explain your reasoning, assuming naphthalene is an aromatic benzene bonded to a diene. [2]

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- (b) When compound **L** reacts with dibenzenesulfonimide via the electrochemical method described in (a), a mixture of isomers, **U** and **V**, was obtained in a 5 : 1 molar ratio.

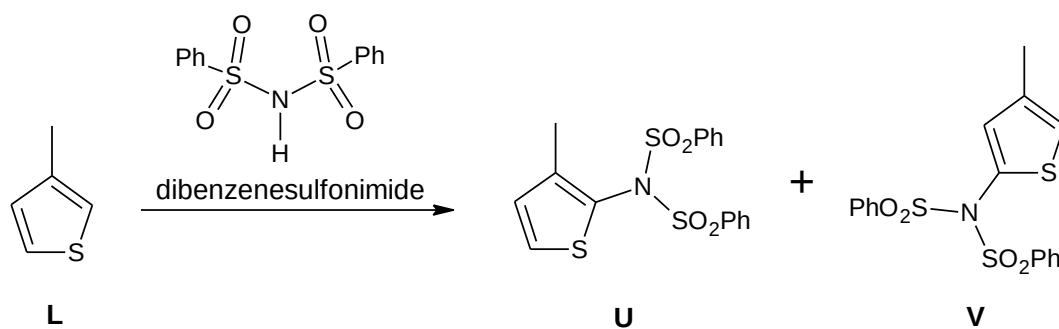


Fig 2.3

Explain the molar ratio of **U**:**V** by drawing structures of the relevant radical intermediates and comparing their relative stabilities. [3]

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(ii) With reference to the diagrams drawn in **(c)(i)**, suggest, with reasons, how

- the bond energy and the bond length of the H–H bond in H_2 molecule would compare with that in the H_2^- ion,
- the first ionisation energy of the H_2 molecule would compare with that of the H_2^- ion.

[2]

[illegible]

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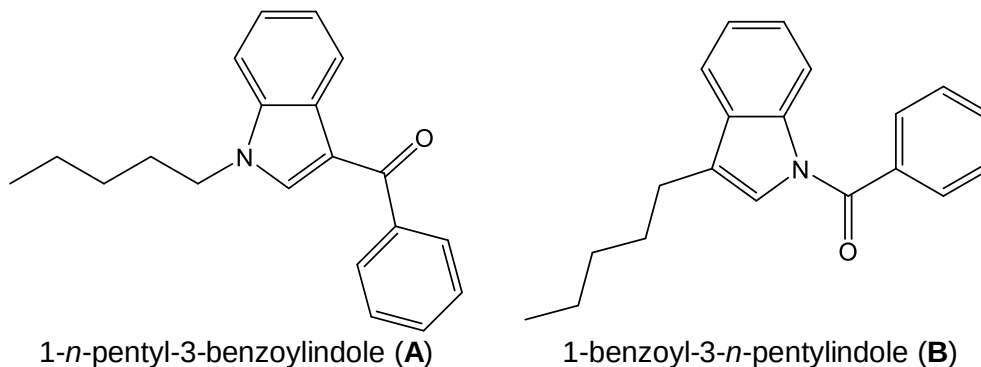


- O=C1NC(=O)c2cc(Cl)cc(Cl)c21

- (i) Outline the basic principles of mass spectrometry. [2]
- (ii) By only considering the isotopes of chlorine and carbon, predict the m/z values of the other five molecular ion peaks, and explain your reasoning. [2]
- (iii) Hence, calculate the relative abundance of each of the five molecular ion peaks. [3]

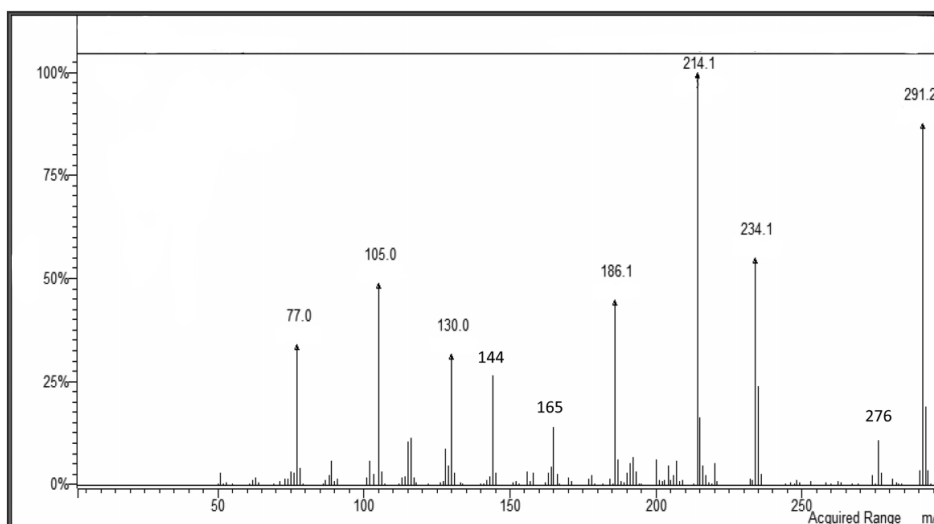
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- (b) A number of indole-derived synthetic cannabinoid have emerged over the past decades in the designer drug market. Examples like 1-*n*-pentyl-3-benzoylindole (**A**) and its regioisomer, 1-benzoyl-3-*n*-pentylindole (**B**), can be prepared using identical precursor reagents and similar reaction conditions.

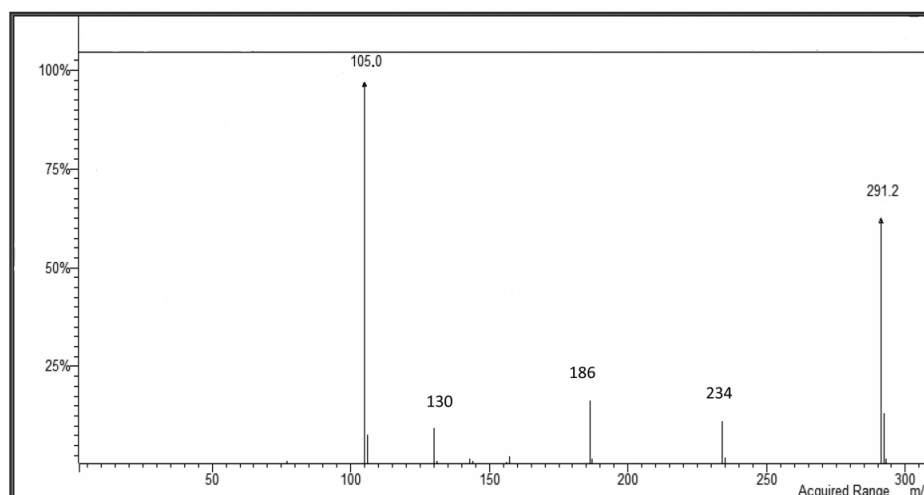


Structural features affect the fragmentation patterns in mass spectra. For instance, cleavage of the amide bond often occurs for aromatic amides.

The mass spectrum of **A** and **B** are shown below.



Spectrum I



Spectrum II

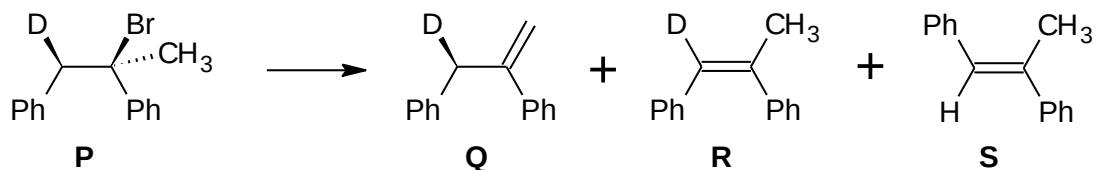
- (i) Suggest the structures of the fragment ions responsible for the base peaks in each spectrum. Account for their high abundance by considering relevant structural features of each ion. [3]
- (ii) Hence deduce which of the spectra I and II correspond with each of the isomers A and B. [1]

[illegible]

[Total: 11]

4 Isotopic labeling is a technique used to track the passage of an isotope through a reaction, making it a useful tool for studying reaction mechanisms.

- (a) Deuterated alkylbromide **P** undergoes elimination to give a mixture of alkenes **Q**, **R** and **S**.



where Ph = phenyl, C₆H₅

The product mixtures obtained under two different reaction conditions are summarised in Table 4.1.

Table 4.1

experiment	reagent used	Q	R	S
I	sodium ethoxide in ethanol	minor product	not formed	major product
II	aqueous ethanol	minor product	minor product	major product

- (i) Assign the stereochemistry of alkene **R**. Explain your reasoning. [1]

- (ii) State a spectroscopic method that will distinguish between **R** and **S**, and describe the measurements needed to identify which is **R** and which is **S**. [1]

Only E2 elimination occurs in experiment I.

- (iii) Draw relevant Newman projections of **P** to explain why **S** is formed but not **R**. [3]

- (iv) Explain why more **S** than **Q** is formed in experiment I. [1]

- (v) Suggest how the conditions in experiment II allow the formation of **R**. [1]

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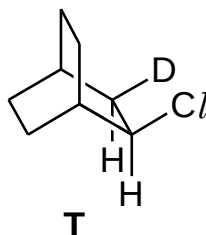
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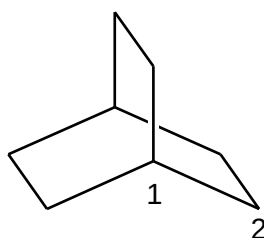
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- (b)** Deuterated alkylchloride **T** undergoes E2 elimination when treated with a strong base like potassium butoxide.

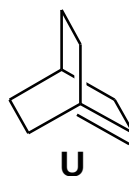


- (i) Predict if the major alkene product contains a deuterium atom. In your answer, you should show the mechanism of the reaction using curly arrows. [2]
- (ii) On the diagram below, draw the orientation of the p-orbitals on C1 and C2 if both atoms were sp^2 -hybridised.



[1]

- (iii) Hence suggest why alkene **U** cannot be formed.



[1]

This image shows a full page of white paper with ten horizontal rows of small black dots, designed for handwriting practice. Each row consists of a single line of evenly spaced dots extending across the width of the page. The rows are separated by equal vertical gaps, providing a guide for letter height and placement. There are no margins, text, or other markings on the page.



Table 4.2

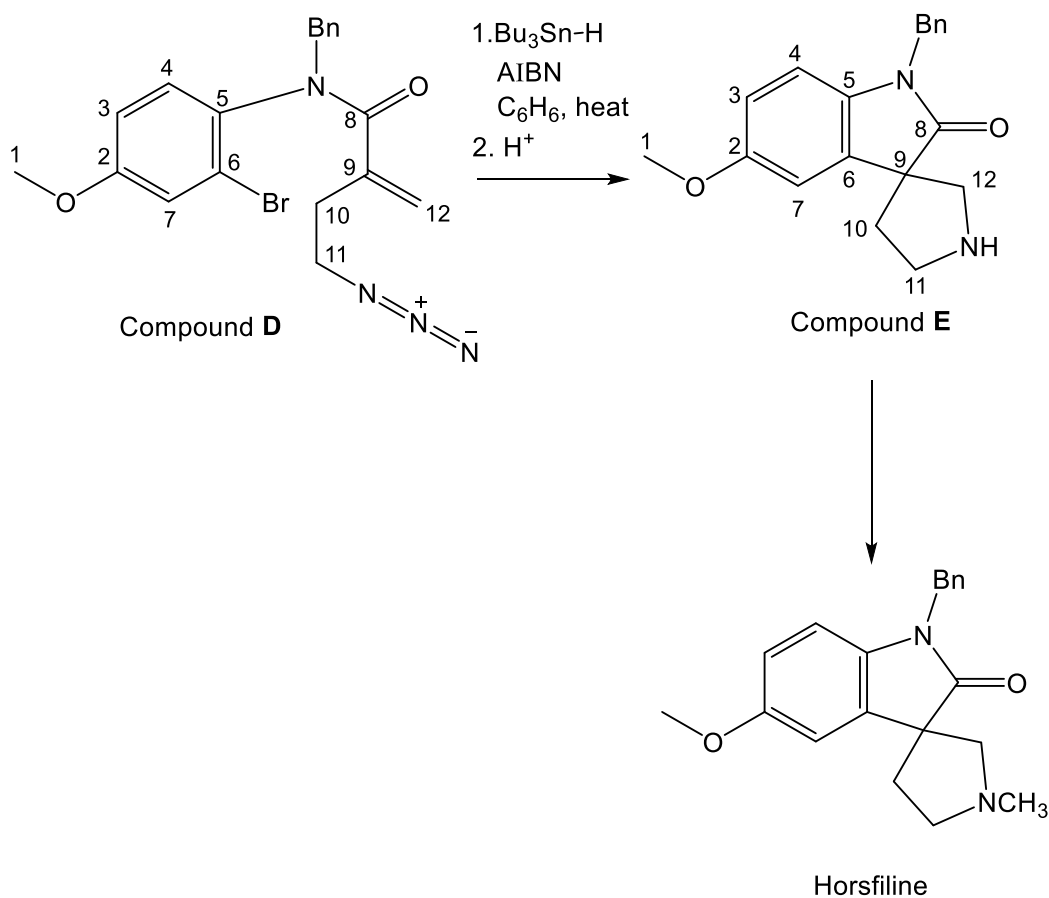
nitrogen base	pK_b
DBN	0.5
trimethylamine	4.2
ammonia	4.7

- (i) Suggest a reason for the low pK_b value of DBN. [1]
- (ii) Give two reasons why the use of DBN instead of ammonia favours elimination over substitution. [2]
- (iii) Besides the choice of base, suggest another change in reaction conditions that can help to favour elimination reaction. Explain your answer. [1]

[illegible]

[Turn Over

- 5 (a) Horsfiline is used in traditional herbal medicine for its analgesic effects. The synthesis of horsfiline follows aryl radical cyclisation.



where Bn = benzyl, $\text{C}_6\text{H}_5\text{CH}_2$ and Bu = butyl, $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2$

The ^1H NMR spectra of **D** and **E** were obtained in the presence of D_2O .

- Explain why the $-\text{NH}$ proton signal is absent from the spectrum of **E**. [1]
- Suggest and explain **two** significant differences you would expect to observe between the spectra of **D** and **E**. [2]
- State and explain the splitting patterns of the peaks at C10 and C12 of **E**. [2]

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- (b) Regioselectivity of electrophilic aromatic substitution of methoxybenzene, $\text{C}_6\text{H}_5\text{OCH}_3$, to produce products with different substitution positions was investigated.

The exact values for ΔH , and the activation energy, E_a , depend on the position of substitution relative to the methoxy group, $-\text{OCH}_3$.

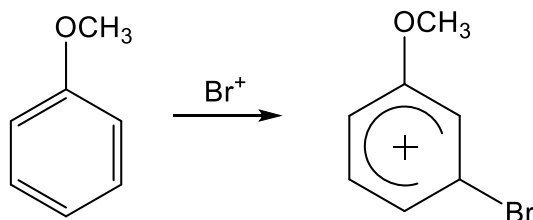
Table 5.1 shows the results when Br^+ electrophile was added to different positions on methoxybenzene.

Table 5.1

Position	$\Delta H/\text{kJ mol}^{-1}$	$E_a/\text{kJ mol}^{-1}$
2 (ortho)	-20.33	13.11
3 (meta)	-3.015	to be calculated
4 (para)	-23.53	10.24

The Bell-Evans-Polanyi Principle relates the activation energies of a series of related reactions to their enthalpy changes.

- (i) Calculate a value for the activation energy of the following reaction, using the data in Table 5.1.



[3]

- (ii) Suggest why the E_a for substitution at the ortho position is higher than that at the para position.

[1]

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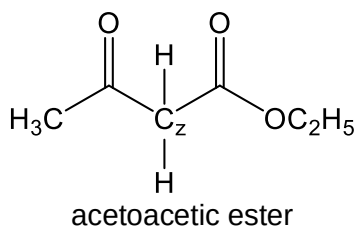
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(b) Acetoacetic ester is the starting material to synthesise **A**.



^1H NMR is a spectroscopic technique used in the characterisation of organic compounds. The ^1H NMR spectrum of acetoacetic ester is shown in Fig. 6.1.

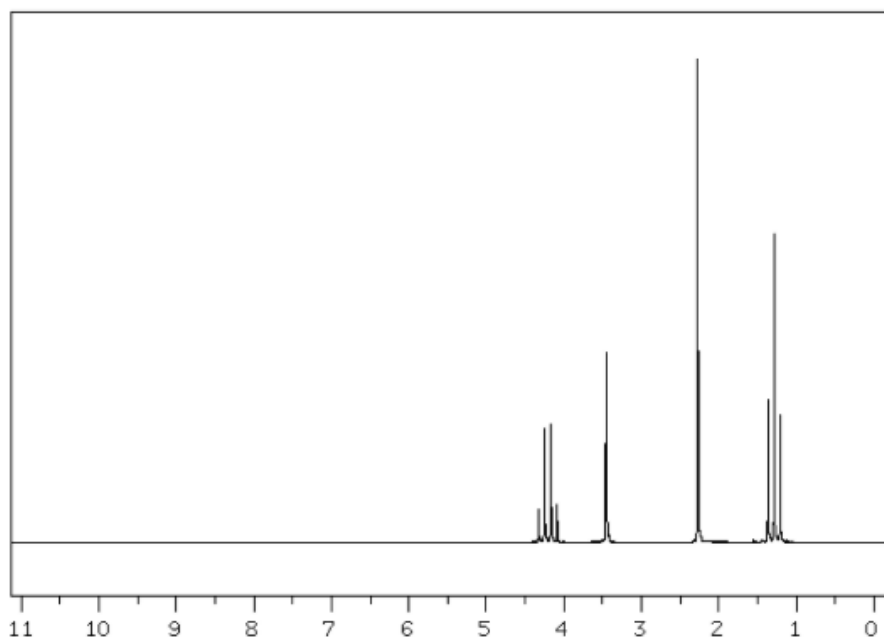


Fig. 6.1

- (i) Assign the peaks in the above spectrum to the appropriate protons in acetoacetic ester and explain your answer. [3]
- (ii) The H atoms on C_z are *relatively easy to deprotonate* in the presence of a strong base like sodium methoxide.

Explain the above statement with the aid of relevant resonance structures. [3]

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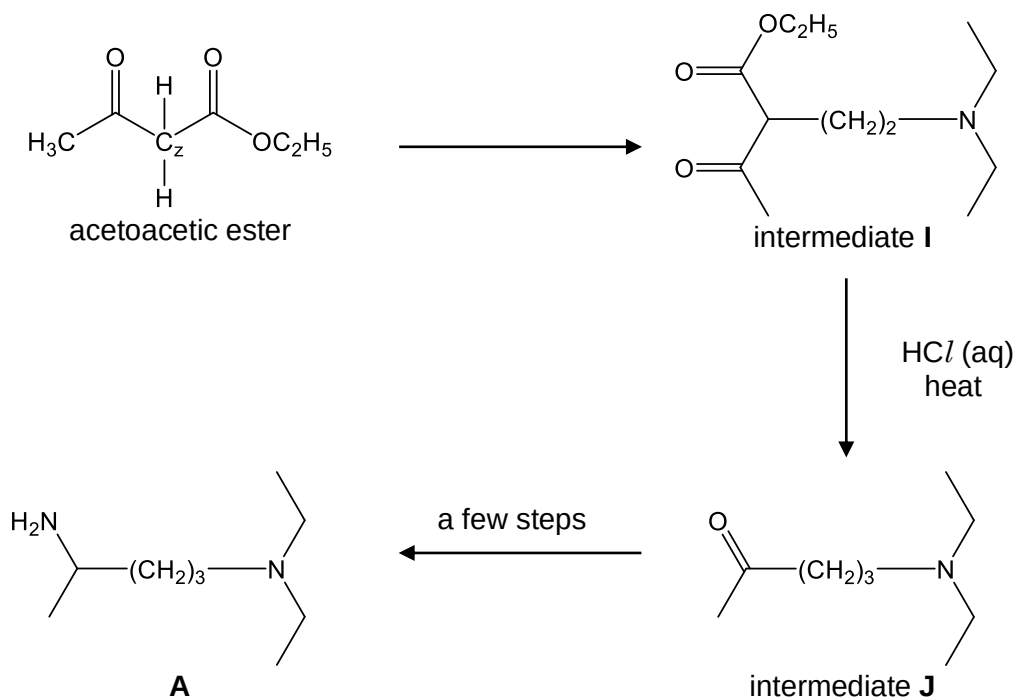
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(c) The actual synthetic pathway of **A** is shown below.



- (i) Using information in (b)(ii), propose the reagents and conditions to form **I** from acetoacetic ester. [1]
- (ii) Propose the reaction pathway, in not more than 3 steps, to form **A** from **J**. State the reagents and conditions clearly. [4]
- (iii) The conversion of **I** to **J** involves acid hydrolysis to give a carboxylic acid functional group, with the subsequent formation of **J** and a by-product via a 6-membered cyclic transition state.

Suggest the mechanism for the formation of enol by indicating curly arrows on Fig 6.2. Hence state the identity of the by-product.

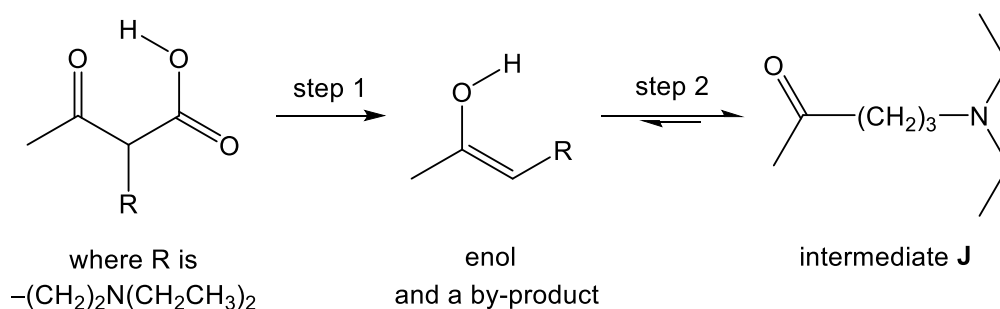
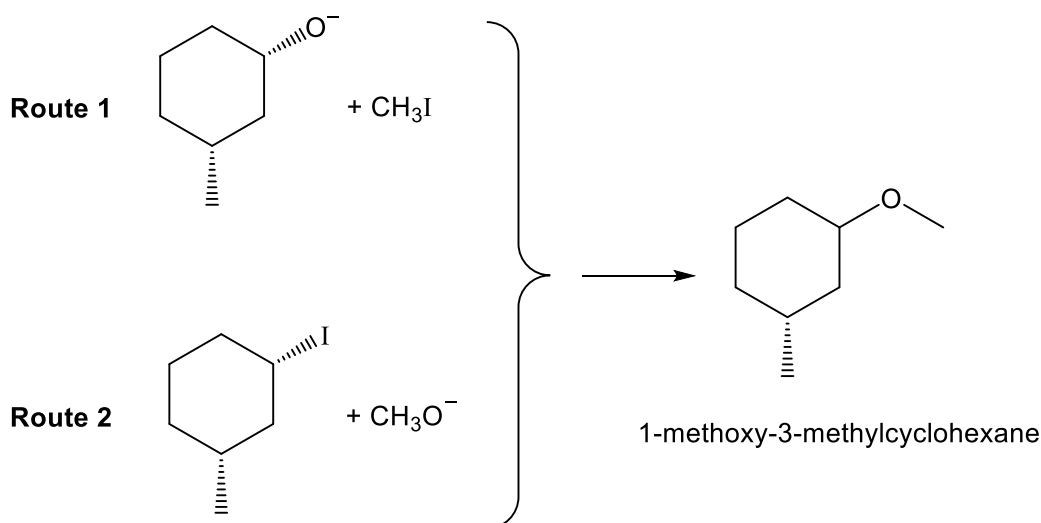


Fig. 6.2

- (iv) Using suitable information from the *Data Booklet*, suggest why the position of equilibrium for step 2 lies well over towards **J**. [2]
- (v) Explain how **I** and **J** could be distinguished using
- infra-red (IR) spectroscopy,
 - ^1H NMR spectroscopy.
- [2]

[Turn Over

- 7 (a) 1-methoxy-3-methylcyclohexane can be prepared from two possible routes shown below. Both routes occur via the bimolecular nucleophilic substitution (S_N2) mechanism.



- (i) Using the *R* and *S* convention, assign the absolute configuration of C1 and C3 for the product formed in **Route 1**. Your answer should include priority assignments as defined by the Cahn-Ingold-Prelog rules. [2]

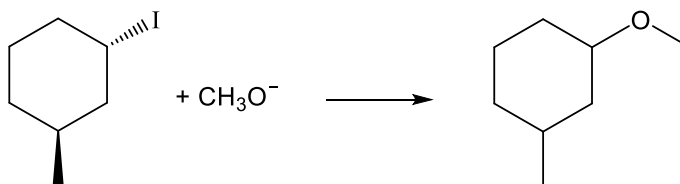
The table below shows the A-values of common substituents on a cyclohexane ring. A-value is the energy difference between the higher energy conformation and the lower energy conformation. It can be used to determine the most stable conformation of cyclohexane rings.

Table 7.1

Substituent	A-value/ kcal mol^{-1}
I	0.43
CH_3	1.70
OCH_3	0.60

- (ii) With the aid of the information in Table 7.1, relevant ring conformations and curly arrows, draw the mechanism for the reaction undergone in **Route 2** to show the stereochemistry of the reaction. In your answer, you should show the more stable conformation of the product. [3]

- (iii) Hence explain why **Route 2** occurs slower than the reaction shown below.



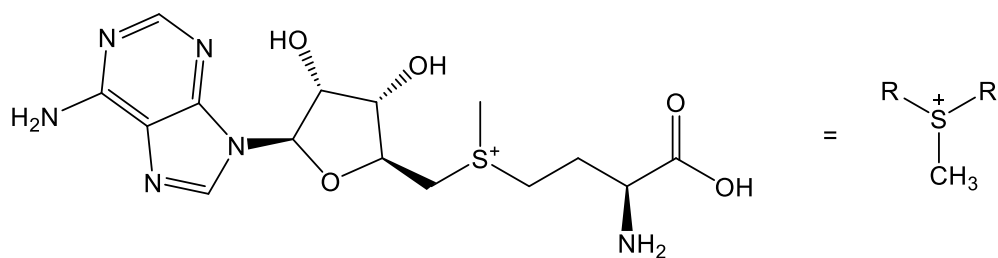
[1]

- (v) It was found that the 1-methoxy-3-methylcyclohexane formed from **Route 1** has a specific rotation of -23.1° and while that formed from **Route 2** has a specific rotation of $+23.1^\circ$.

Determine the optical purity and percentage composition of a mixture that contains both compounds whose specific rotation was found to be $+18.4^\circ$, giving your answers to the nearest whole number. [2]

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- (b) One important biological reaction is *N*-methylation. It is a nucleophilic substitution reaction catalysed by *S*-adenosylmethionine (SAM). SAM is a type of methyltransferase enzyme that plays the role of methyl group donor.



S-adenosylmethionine (SAM)

In a bacterial DNA, *N*-methylation involving SAM and an adenine residue occurs in the presence of an enzyme. The first step of the mechanism is shown in Fig. 7.1.

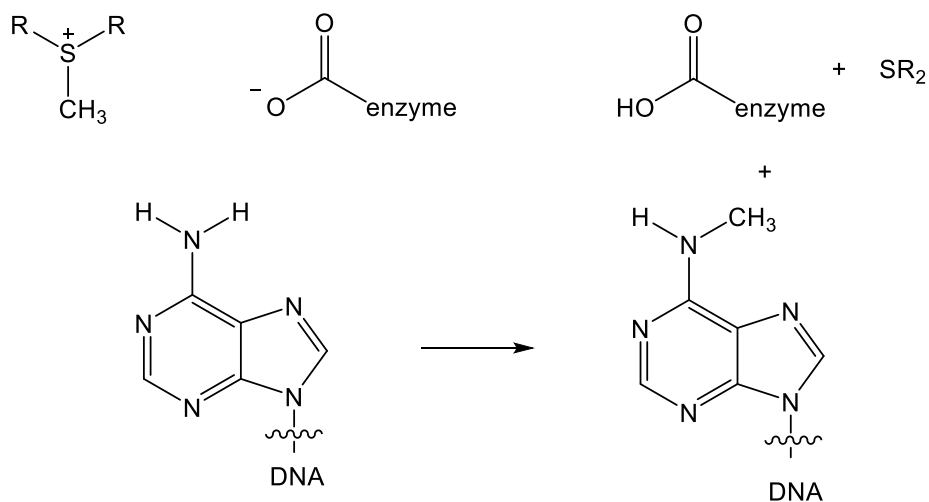


Fig. 7.1

- (i) It was proposed that the reaction takes place in a concerted fashion. Complete the mechanism on Fig. 7.1 by drawing appropriate curly arrows. [1]

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- (c) (i) Explain the origin of infra-red (IR) absorptions of simple molecules. [2]
- (ii) Suggest two reasons why the IR absorption frequency of the C=O group in carbon dioxide (2350 cm^{-1}) is greater than that in propanone (1730 cm^{-1}). [3]
- (iii) Suggest a factor which needs to be considered if IR spectroscopy was to be used as a way of monitoring the progress of a reaction. [1]

[illegible]

[Turn Over

- 8 (a) α,β -unsaturated carbonyl compounds like acrolein have different physical and chemical properties compared to isolated alkenes and carbonyl compounds. Most notably, these compounds are susceptible to nucleophilic attack at the β carbon position.

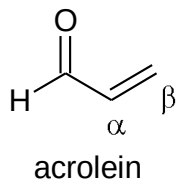


Table 8.1 lists the ultraviolet (UV) wavelength of maximum absorbance of propene, propanone and acrolein.

Table 8.1

compound	Wavelength of maximum absorbance, λ_{max} / nm
propene	176
propanone	274
acrolein	322

State the type of electronic transition that is responsible for each of these UV absorptions and explain the difference observed. [3]

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- (b) Figure 8.1 shows how enamine **A** undergoes 1,4-addition with acrylonitrile to give two different products at different reaction temperatures.

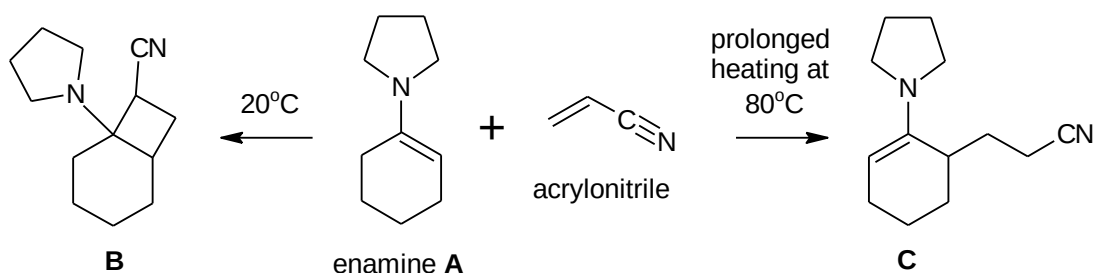


Fig. 8.1

Hence explain why **B** and **C** are the major products of the reactions carried out in Fig. 8.1 under different conditions. In your answer, you should consider the activation energy and reversibility of each reaction. [3]

- (c) The Skraup reaction is a method for the one-pot synthesis of quinolines. A mixture containing a phenylamine, an α,β -unsaturated carbonyl, sulfuric acid, and an oxidising agent are heated together. A sequence of reactions starting with a 1,4-addition occurs in the reaction flask as shown in Fig. 8.2.

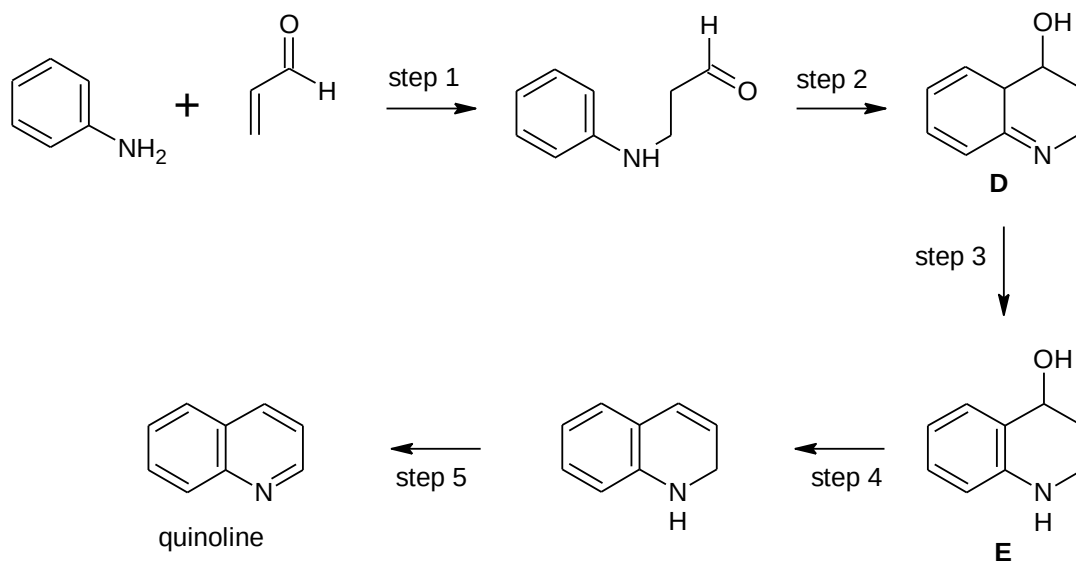


Fig. 8.2

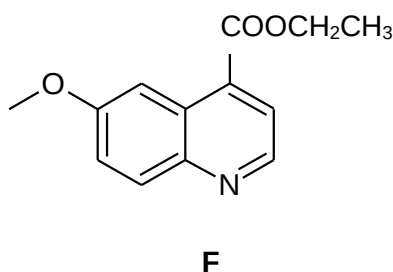
- (i) Suggest one advantage of a one-pot synthesis. [1]
- (ii) By considering the oxidation number of the reacting carbon, show that the oxidising agent is used in step 5. [1]
- (iii) Suggest the role of sulfuric acid in the synthesis. [1]

The conversion of **D** to **E** in step 3 occurs readily as the product is lower in energy.

- (iv) Considering the π systems only, sketch the combination of p-orbitals in the Highest Occupied Molecular Orbitals (HOMO) of **D** and **E**. You can treat **D** as hexatriene and **E** as benzene. You may ignore the effects of a heteroatom on relative energy levels.

Hence explain the relative stabilities of **D** and **E**. [3]

Quinoline **F** is used to synthesise quinine, an antimalarial drug.



- (v) Identify the phenylamine and α,β -unsaturated carbonyl that can be used to synthesise **F** using the Skraup reaction. [2]

- (d) The concentration of quinine in a 320 ml can of tonic water is determined using UV spectroscopy. Quinine has an absorption peak at 345 nm with a molar extinction coefficient, $\epsilon = 5710 \text{ mol}^{-1} \text{ dm}^3 \text{ cm}^{-1}$. At this wavelength, the absorbance of the tonic water sample in a cell of path length 1 cm was 1.18.

(i) Calculate the concentration of quinine in the tonic water sample. [1]

(ii) The effective dose of quinine for antimalarial effect is 8 mg per kg of bodyweight.

Using your answer to (d)(i), determine the total number of cans of tonic water an average 70 kg adult will need to ingest for malarial effect.

[M_r of quinine = 324.4] [2]

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